Characterization of conserved Flavivirus 5'UTR elements

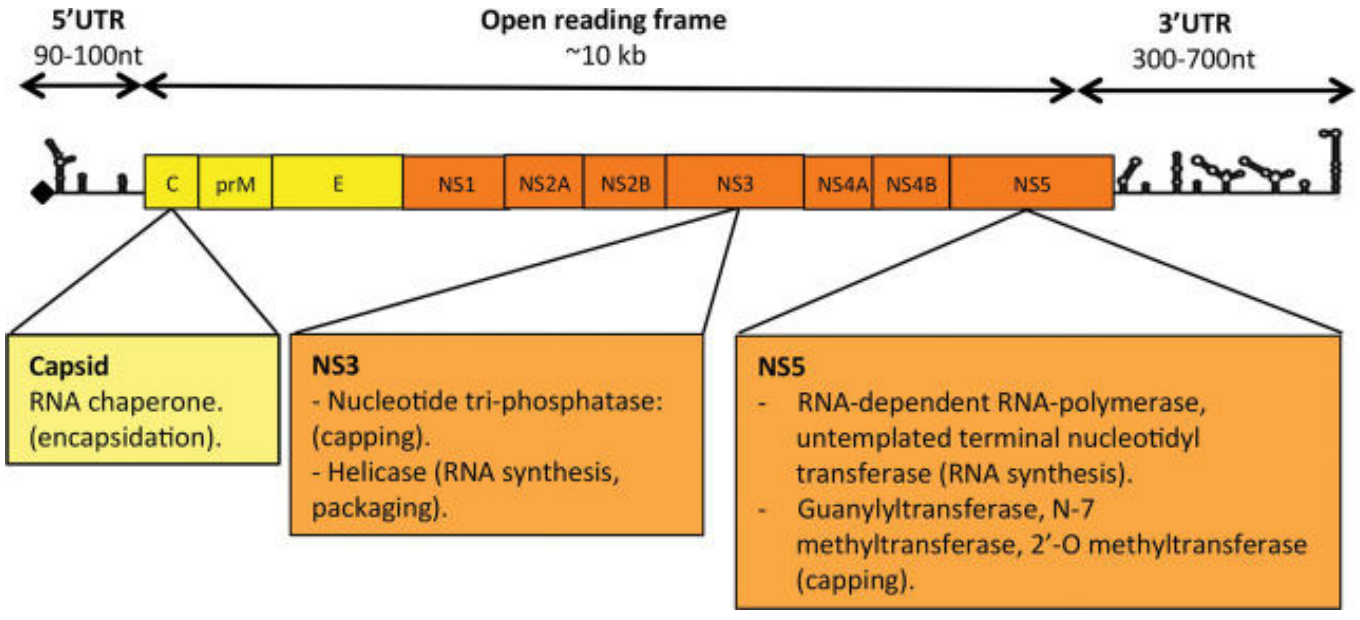
Michael T. Wolfinger

Department of Theoretical Chemistry
University of Vienna

Bled, Slovenia, February 16, 2017

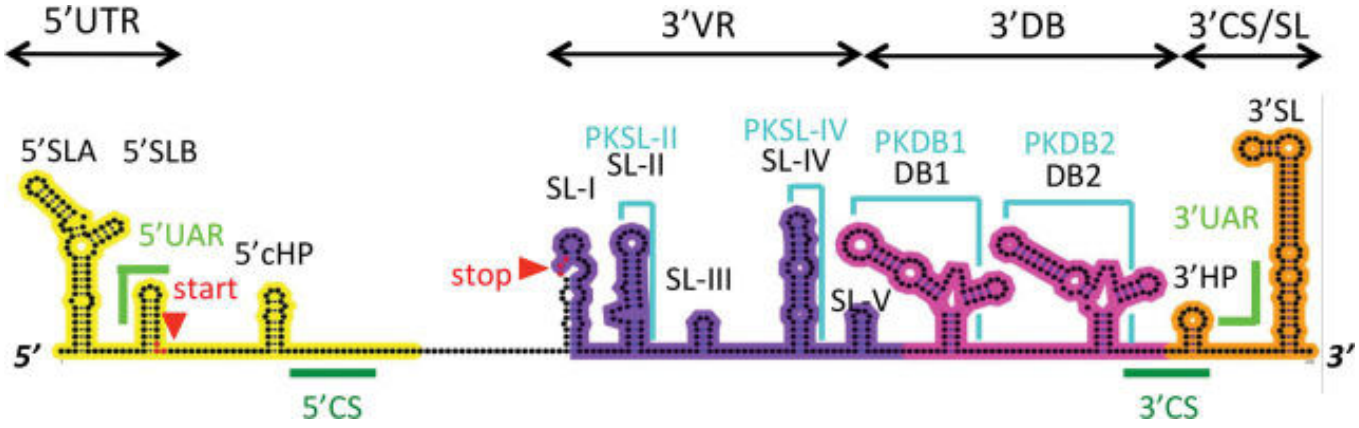
Flavivirus genome characteristics

- Single-stranded, positive-strand RNA viruses
- Enters cell through receptor-mediated endocytosis
- Capped, non-polyadenylated genome (gRNA) of 10-12kb length
- Encodes a single ORF, flanked by structured 5'-UTR and 3'-UTR
- Translation of FV ORF yields a single polyprotein



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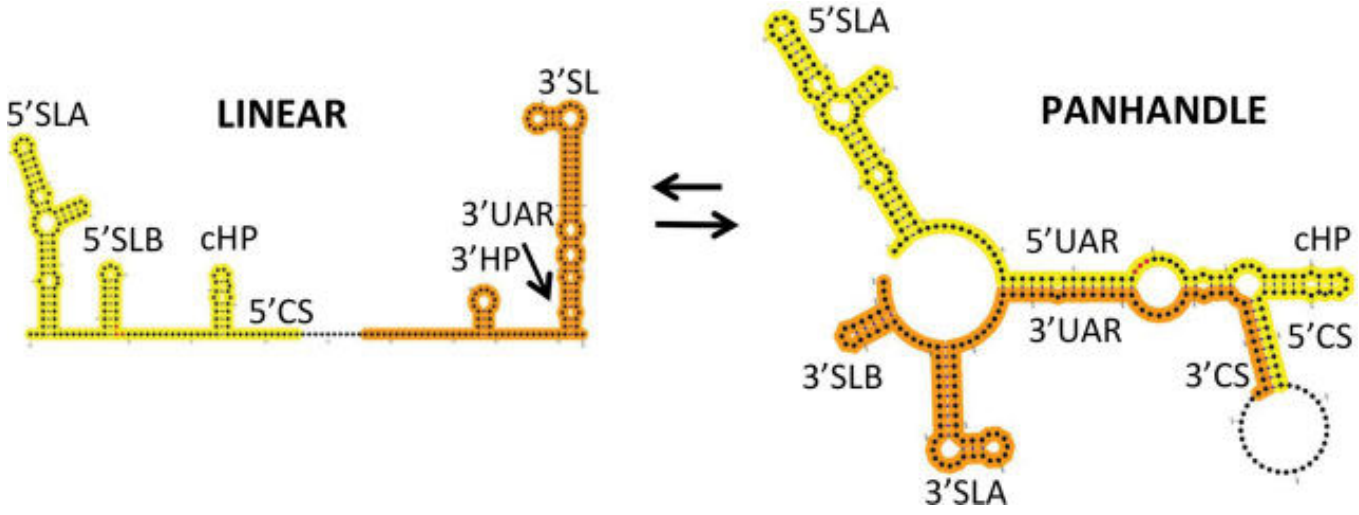


Flavivirus replication

- FV RNA replication happens via asymmetric replication cycle
- Viral RNA-dependent RNA polymerase (RdRp) synthesizes a full-length minus-strand (-)gRNA
- (-)gRNA is a template for several rounds of (+)gRNA synthesis
- Ratio (-)gRNA / (+)gRNA between 1:10 and 1:100
- Genome cyclisation is required for (-)gRNA synthesis

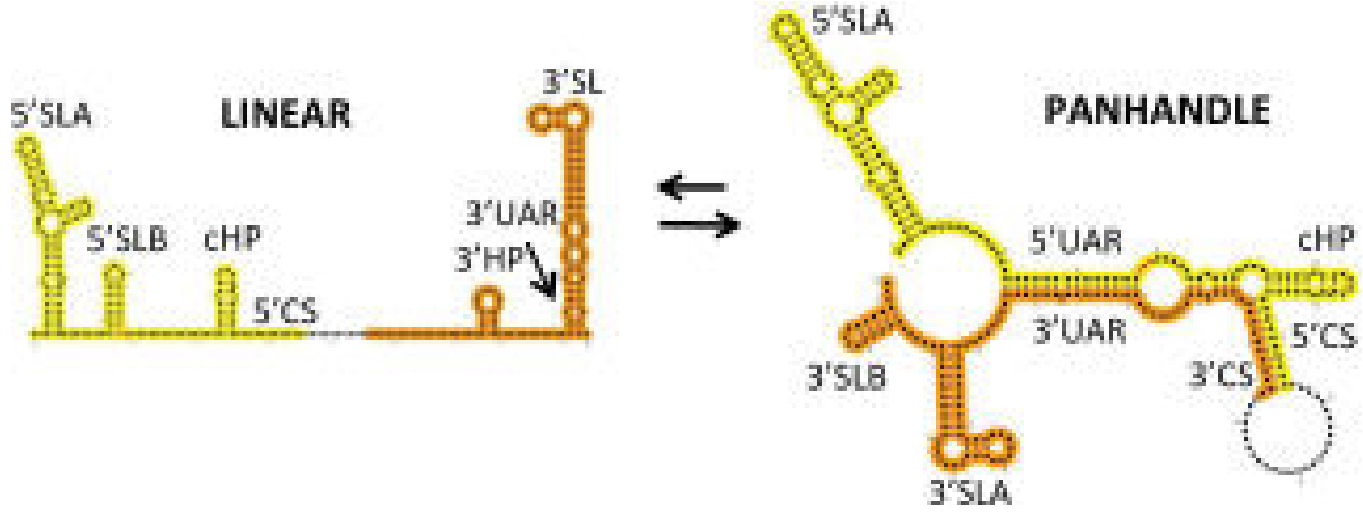
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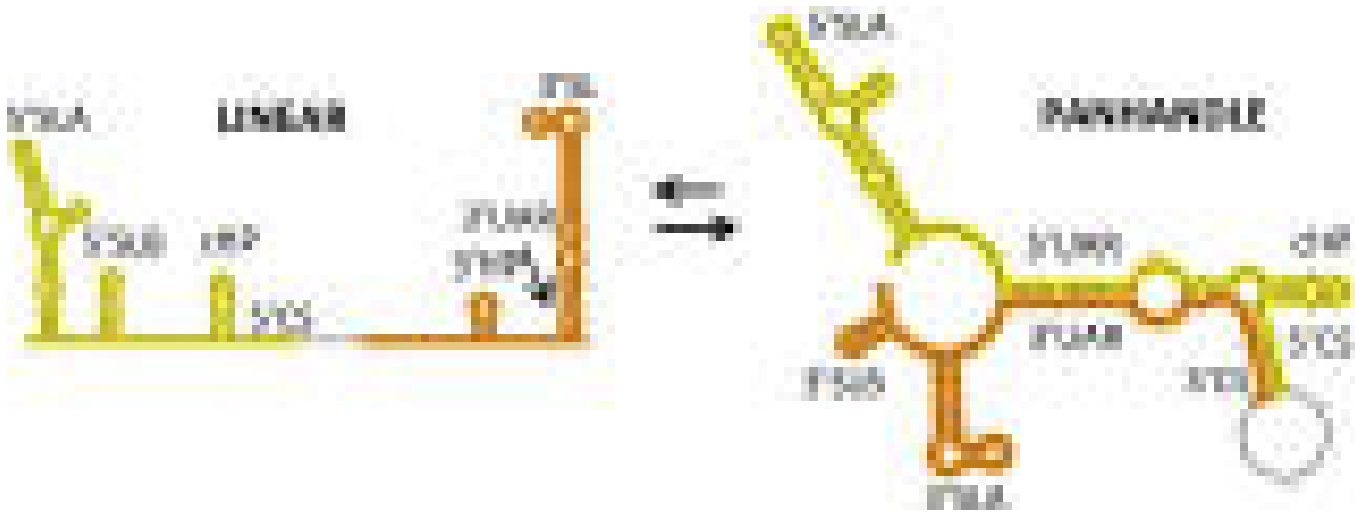
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- SLA acts as promoter / enhancer for viral RdRp

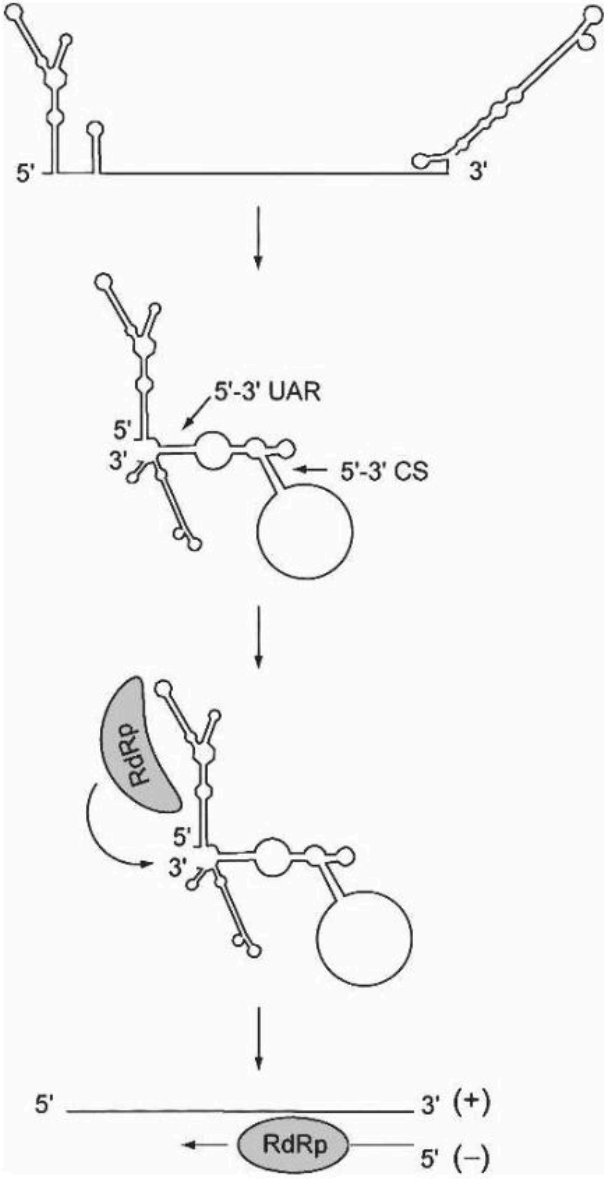
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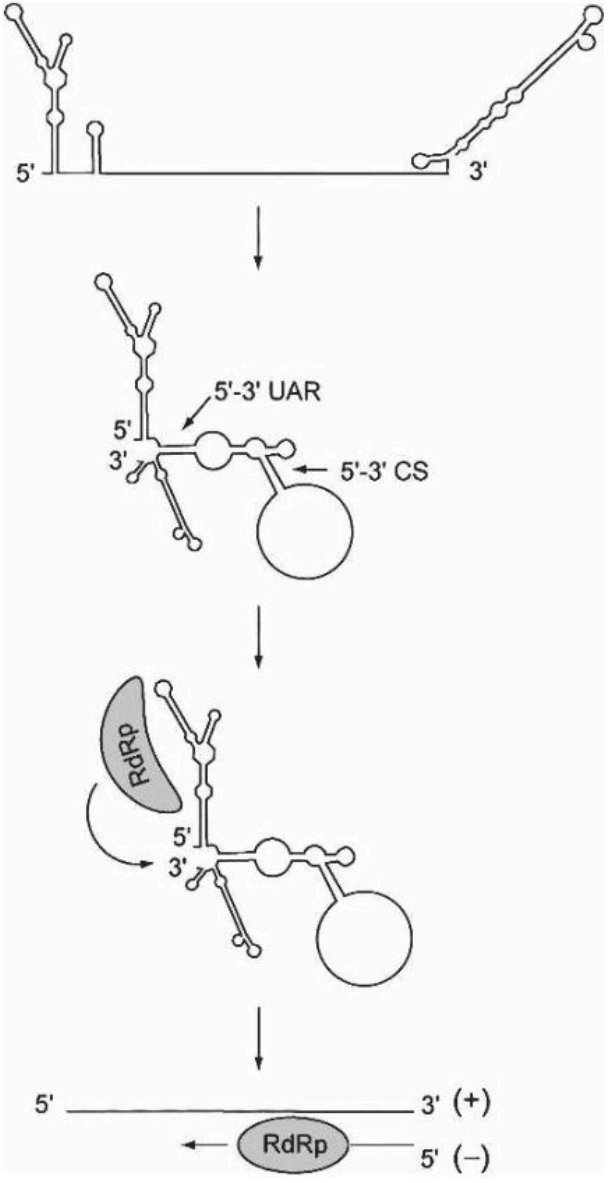
- SLA acts as promoter / enhancer for viral RdRp
- Viral RdRps are error-prone: Misincorporations every 40-70kb

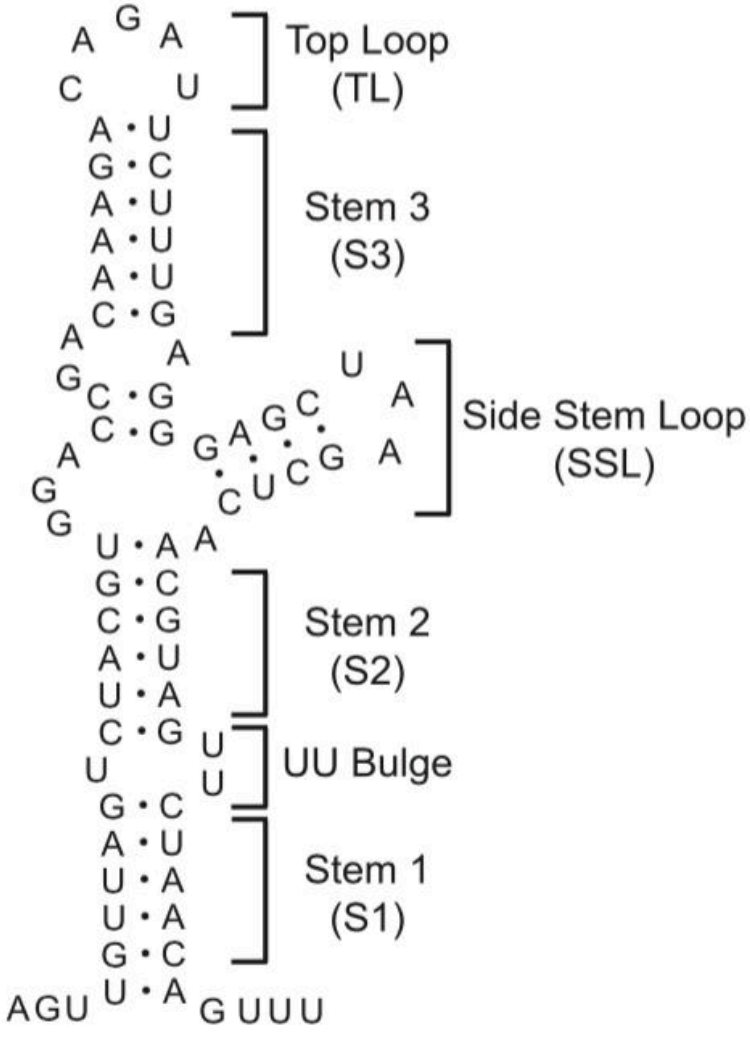
Flavivirus replication



Lodeiro et al., J. Virol. (2009)

Flavivirus replication



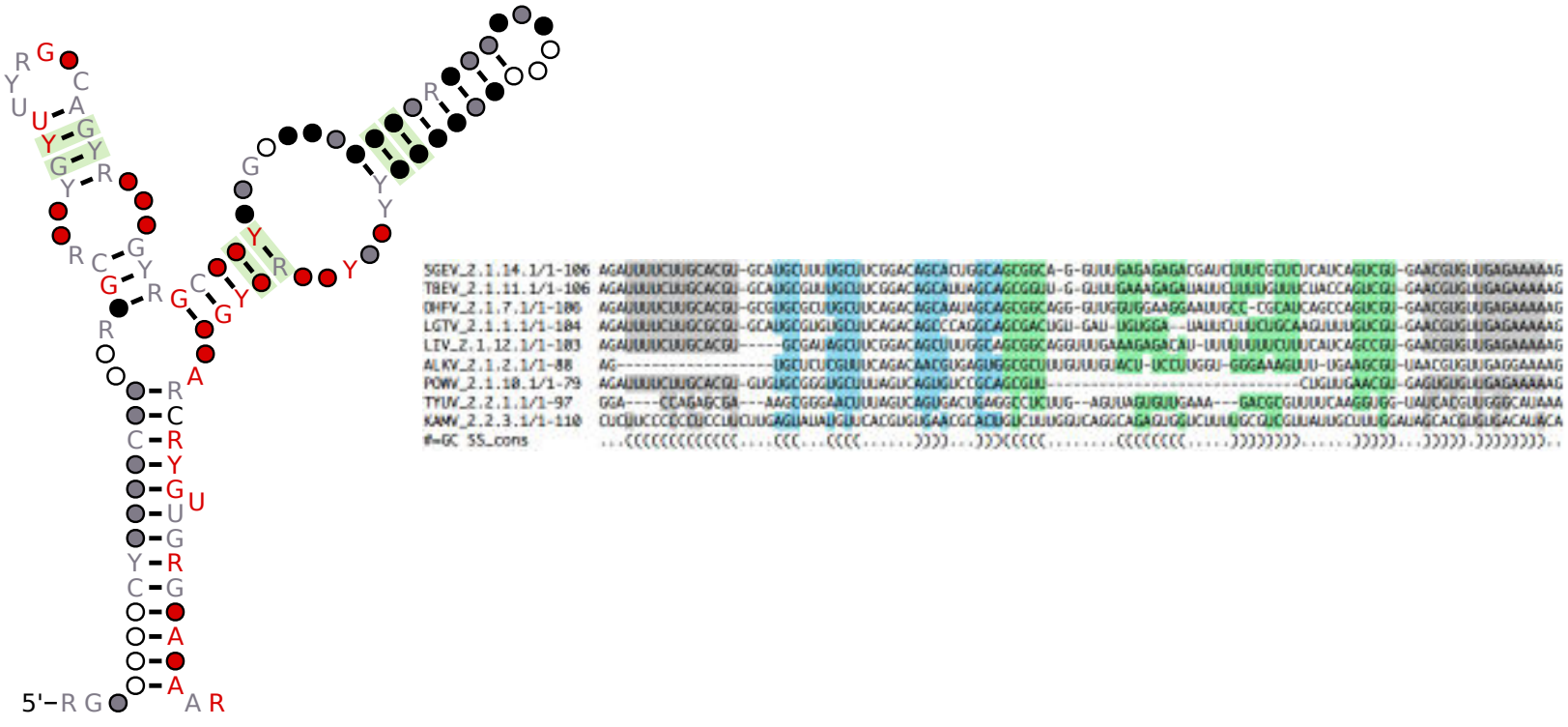


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SLA in different FV groups

Consensus structures computed from mlocarna alignments of all non-redundant representatives for each group obtained from NCBI

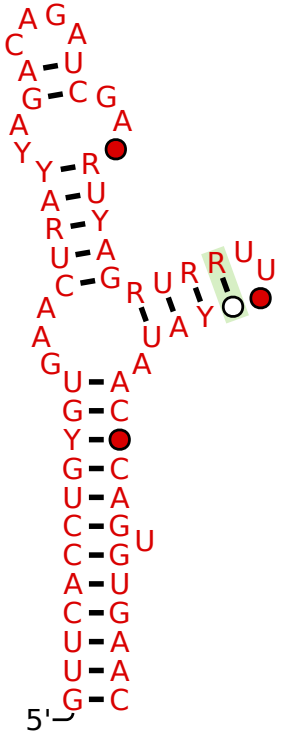
Tick-bourne Flavivirus group (TBFVG)



SLA in different FV groups

Consensus structures computed from mlocarna alignments of all non-redundant representatives for each group obtained from NCBI

Kokobera virus group (KOKVG)

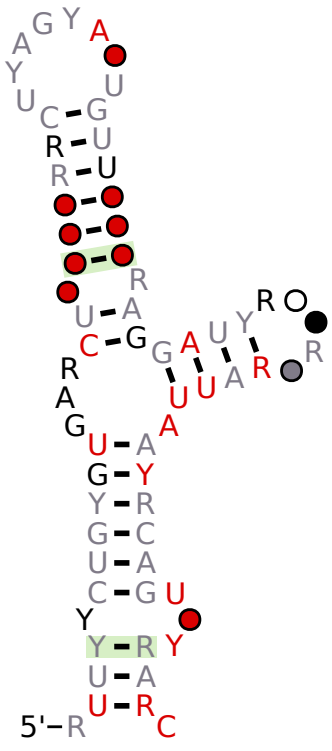


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NMV.1.4.2.1.1.1     GUUCACCUGCGUGAACUGAUUAGACAGAUCGACAUCAGAUAGUUUCUAUAACUCAGUGUGAAC
#=GC SS_cons        ((((((((((((...(((((..((....))...)))))(((...)))).))))))).))))))
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SLA in different FV groups

Consensus structures computed from mlocarna alignments of all non-redundant representatives for each group obtained from NCBI

Japanese encephalitis virus group (JEVG)

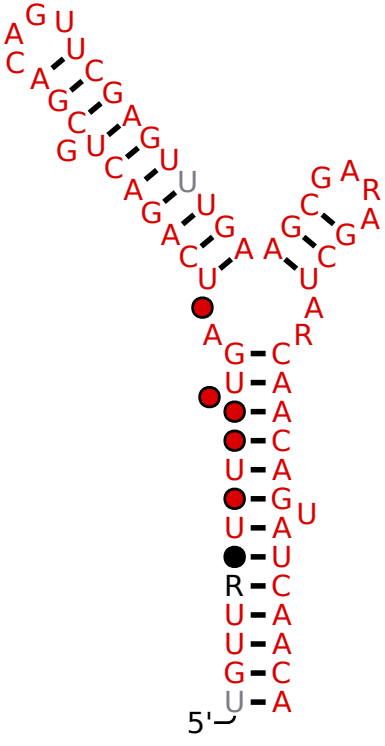


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JEV.1.3.1.163.1.1  G-UUA--AUCUGUGUGA---ACUUCUUGGCUUAGUAUCGU-UGAGAAGAAUCG-AGA-GAUUAGU-GCAGUUUAAAC-
JEV.1.3.1.130.1.1  G-UUU--AUCUGUGUGA---ACUUCUUGGCUUUGUAUCGU-UGAGAAGAAUCG-AGA-GAUUAGU-GCAGUUUAAAC-
ALFV.1.3.5.1.1.1   G-UUC--AUCUGUGUGA---ACUUUUUGACUCAGGAUUGU-UGGAAGGGAUUG-AAA-GAUUAAU-ACAGUUUGAAC-
JEV.1.3.1.165.1.1  G-UUU--AUCUGUGUGA---ACUUCUUGGUUUAGUAUCGU-UGAGAAGAAUCG-AAA-GAUUAGU-GCAGUUUAAAC-
JEV.1.3.1.75.1.1   G-UUU--UAACGUGUGA---ACUUCUUGGCUUAGUAUCGU-CGAGAGAGAUCG-AGA-GAUUAGU-GCAGUUUAAAC-
JEV.1.3.1.82.1.1   G-UUUU-UUUUGUGUG----ACUUCUGGGCUUAGUAUCGU-UGAGAAGAAUCG-AGA-GAUUAGU-GCAGUUUAAAC-
JEV.1.3.1.84.1.1   G-UUUU-AACUGUGUG----ACUUCUUGGCUUAGUAUCGU-UGAGAAGAAUCG-AGA-GAUUAGU-GCAGUUUAAAC-
USUV.1.3.7.89.1.1  G-UUG--GCCUGUGUGA---GCUCUACUACUUAGUAUUGU-UUUGGAGGAUCG-UUA-GAUUAAU-ACAGUGUUGAC-
MVEV.1.3.6.2.1.1   G-UUC--AUCUGCGUGA---GCUUCCGAUCUCAGUAUUGU-UUGGAAGGAUCA--UU-GAUUAAC-GCGGUUUGAAC-
MVEV.1.3.6.1.1.1   G-UUC--AUCUGCGUGA---GCUUCCGAUCUCAGUAUUGU-UUGGAAGGAUCA--UU-GAUUAAC-GUGGUUUGAAC-
MVEV.1.3.6.4.1.1   G-UUC--AUCUGCGUGA---GCUUCCGAUCUCAGUAUUGU-UUGGGAGGAUCA-UUU-GAUUAAC-GCGGUUUGAAC-
MVEV.1.3.6.9.1.1   G-UUC--AUCUGCGUGA---GCUUUCGAUCUCAGUAUUGU-UUGAAAGGAUCAUUUU-GAUUAAC-GCGGUUUGAAC-
WNV.1.3.2.1311.1.1 GUUUC--GCCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUAACAACAAUUAAC-ACAGUGCGAGC-
WNV.1.3.2.1370.1.1 GUCUC-AGCCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUAACAACAAUUAAC-ACAGUGCGAGC-
WNV.1.3.2.1371.1.1 GUAUUGACCCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUAACAACAAUUAAC-ACAGUGCGAGC-
WNV.1.3.2.1479.1.1 --GUCG--CCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUAACAACAAUUAAC-ACAGUGCGAGC-
WNV.1.3.2.973.1.1  G-UUCG--CCUGUGUGA---ACUGACAAACUUAGUAGUGU-UUGUGAGGAUUAAUAACAAUUAAC-ACAGUGCGAGC-
KUNV.1.3.4.43.1.1  G-UUCG--CCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUUUGAACAAUUAAC-ACAGUGCGAGC-
WNV.1.3.2.1356.1.1 ACAAC---CCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUAACAACAAUUAAC-ACAGUGCGAGC-
WNV.1.3.2.1683.1.1 --UUCG--CCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUUACGAGGAUUAAU-ACAGUACGAGC-
WNV.1.3.2.1408.1.1 G-------CCUGUGUGA---GCUGACAAACUUAGUAGUGU-UUGUGAGGAUUAACAACAAUUAAC-ACAGUGCGAGC-
USUV.1.3.7.10.1.1  GUUCG---UCUGGGUGA---GCUCUACUACUUAGUAUUGUUUUUGGAGGAUCGUGA--GAUUAAC-ACAGUGCCGGC-
USUV.1.3.7.95.1.1  GUUCG---UCUGCGUGA---GCUCUACUACUUAGUAUUGUUUUUGGAGGAUCGUGA--GAUUAAC-ACAGUGCCGGC-
USUV.1.3.7.24.1.1  GUCCG---UCUGCGUGA---GCUCUACUACUUAGUAUUGUUUUUGGAGGAUCGUGA--GAUUAAC-ACAGUGCCGGC-
USUV.1.3.7.46.1.1  GUUCG---CCUGCGUGA---GCUCUACUACUUAGUAUUGUUUUUGGAGGAUCGUGA--GAUUAAC-ACAGUGCCGGC-
USUV.1.3.7.9.1.1   GUUCG---UCGGCGUGA---GCUCUACUACUUAGUAUUGUUUUUGGAGGAUCGUGA--GAUUAAC-ACAGUGCCGGC-
SLEV.1.3.3.28.1.1  UGUUCG--CGUCGGUGA---GCGAAGAGGAAACAGAUUUC-UUUUUUGGAGGAUAACAACUUAACUUGAUUGCGAACA
SLEV.1.3.3.30.1.1  UGUUCG--CGUUGGUGA---GCGAGGAAGAAACAGAUUUC-UUUUUUGGAGGAUUACAACUUAACUUGACUGCGAACA
WNV.1.3.2.1616.1.1 UCUCC---UCAGAAUC-----CUGAAAAGAUUCUCAGUGU-UUGUGAGGAUUAACAACAAUUAAC-ACAGUGCGAGC-
#=GC SS_cons       ..(((....((((((......((.((((((........)).)))).))((((......))))..)).))))...)))..
```

SLA in different FV groups

Consensus structures computed from mlocarna alignments of all non-redundant representatives for each group obtained from NCBI

Spondweni virus group (SPOVG)

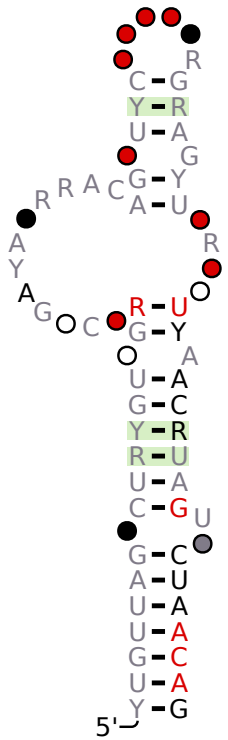


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ZIKV.1.6.2.21.1.1   UGUUGAUCUGUGUGAGUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.161.1.1  UGUUGAUCUGUGUGAAUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.158.1.1  UGUUGAUCUGUGUGAGUCAGACUGCGACAGUUCGAGUCUGAAGCGAGAGCUAACAACAGUAUCAACA
ZIKV.1.6.2.169.1.1  UGUUGAUCUGUGUGAAUCAGACUGCGACAGUUCGAGUUUGAAGCGAGAGCUAACAACAGUAUCAACA
ZIKV.1.6.2.84.1.1   AGUUGAUCUGUGUGAAUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.150.1.1  UGUU--UCUGUGUGAAUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.141.1.1  UGUUACUGUUGCUGACUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
#=GC SS_cons        (((((((((((.((..(((((((.(((....)))))))))))(((....))).))))))).)))))))
```

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Dengue virus group (DENVG)

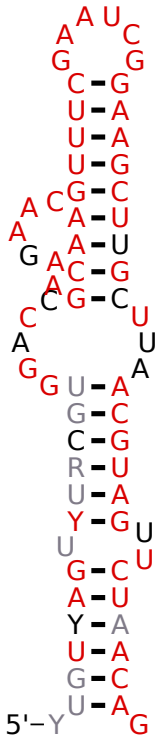


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DENV2.1.2.2.51.1.1    CUG---UUAG-U-CUACGUUGAGCUUAGCUCCC-UCAAAGAAUCUGUCUUUGUCGG---------UCCACGUAGA-CUAACAGUU
DENV2.1.2.2.52.1.1    CUG---UUAGAA-CUACGUUGAGCUUAGCUCCC-UCAAAGAAUCUG-UCUUUGAGGGAGCUAAGCUCAACGUAGUUCUAACAG--
DENV4.1.2.4.11.1.1    CUG---UUAGAA-CUGUGUUAAG--------CA-AGCUUCCGAUU-UGGAAGCUUGG--------UCCACACAGUUCUAACAG--
DENV1.1.2.1.237.1.1   GCG-------CUUCCGAUCUGGACCGACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---------UUAACGUAGUUCUAACAG--
DENV1.1.2.1.234.1.1   AUG---------UCC---UUGGACCGACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---------UUAACGUAGUUCUAACAG--
DENV1.1.2.1.311.1.1   UUGU--AAAACGGACGCCAGUGGCCGACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---------UUAACGUAGUUCUAACAG--
DENV4.1.2.4.66.1.1    GUG---UUUGUG-UUGUGUGGACC-GACAAGGA-CAGUUCCAAAUCGGAAGCUUGC---------UUAACACAGUUCUAACAG--
DENV4.1.2.4.65.1.1    UUG---UUUAGG-UUGUGUGGACC-GACAAGGA-CAGUUCCAAAUCGGAAGCUUGC---------UUAACACAGUUCUAACAG--
DENV3.1.2.3.251.1.1   UUG---UUAG-U-UUUCGUGGACC-GACAAGAA-CAGUUUCGACUCGGAAGCUUGC---------UUAACGUAGUGCUGACAG--
DENV3.1.2.3.199.1.1   UUG---UUAG-U-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---------UUAACGUAGUGCUAACAG--
DENV3.1.2.3.800.1.1   UUG---UUAG-U-CUACGUGGACC-GACAAGAA-CAGUUUCGACUCGGAAGCUUGC---------UUAACGUAGUGCUAACAG--
DENV3.1.2.3.11.1.1    UUG---UUAG-U-CUACGUGGACC-GACAAGAA-CAGUUUCAACUCGGAAGCUUGC---------UUAACGUAGUGCUGACAG--
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DENV1.1.2.1.402.1.1   GAC---UCAG-A-UACCACGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---------UUAACGUAGUUCUAACAG--
DENV2.1.2.2.386.1.1   AAG------U-C-CUACGU-GGACCGACAAAGA-CAGAUUCUUUGAGGGAGCUAAGC--------UCAACGUAGUUCUAACAG--
DENV2.1.2.2.367.1.1   UUC------U-C-CUACGUUGGACCGACAAAGA-CAGAUUCUUUGAGGGAGCUAAGC--------UCAACGUAGUUCUAACAG--
DENV2.1.2.2.256.1.1   U-----------CUACGU-GGACCGACAAAGA-CAGAUUCUUUGAGGGAGCUAAGC--------UCAACGUAGUUCUAACAG--
DENV2.1.2.2.365.1.1   CAG---AGAC-A-CUACGU-GGACCGACAAAGA-CAGAUUCUUUGAGGGAGCUAAGC--------UCAACGUAGUUCUAACAG--
DENV2.1.2.2.372.1.1   GUC---UCAG-C-UAACGU-GGACCGACAAAGA-CAGAUUCUUUGAGGGAGCUAAGC--------UCAACGUAGUUCUAACAG--
DENV2.1.2.2.276.1.1   GGG---GCAG---UUACGU-GACCCGACAAAGA-CAGA-UCUUUGAGGGAGCUAAGC--------UCAACGUAGUUCUAACAG--
KEDV.1.2.5.1.1.1      UU----UUGA-U-CG-GUGUGAAUCGAUACAAA-CAGAUUUUGUAGAGAAGUUAUUC--------UAACACUCGUGUCAACA---
DENV2.1.2.2.389.1.1   GACUCAGACUCUCCUACGU-GGACCGACAAAAGACAGAUUCUUUGAGGGAGCUAAGC--------UCAACGUAGUUCUAACAG--
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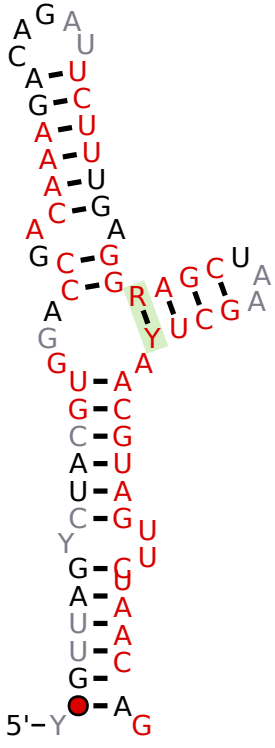
Dengue virus SLA

There are four DENV serotypes, probably five ...

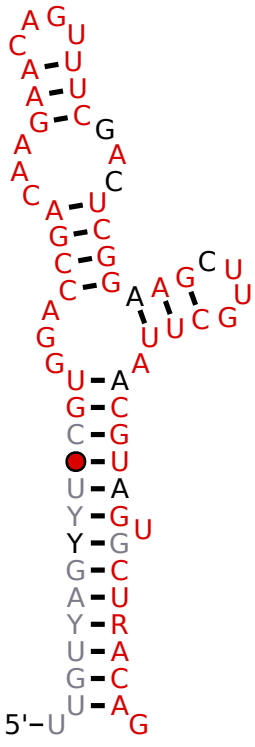
DENV1



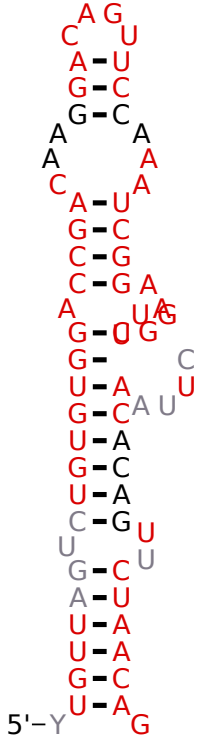
DENV2



DENV3



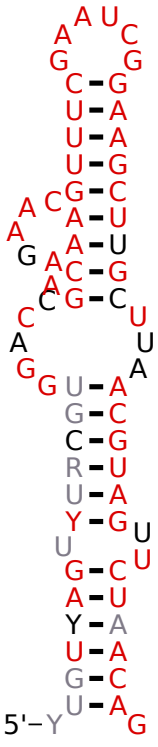
DENV4



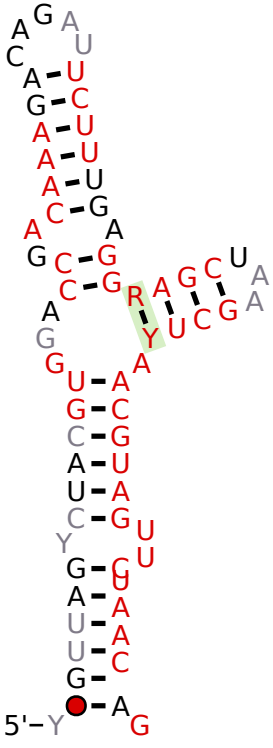
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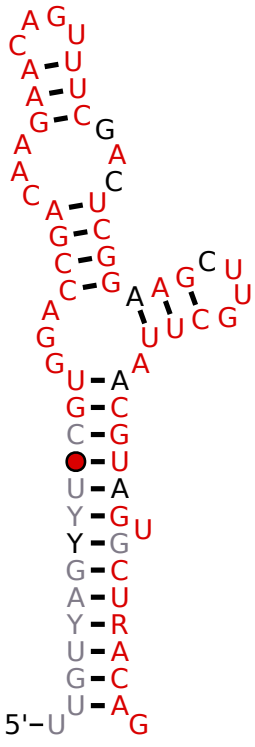
DENV1



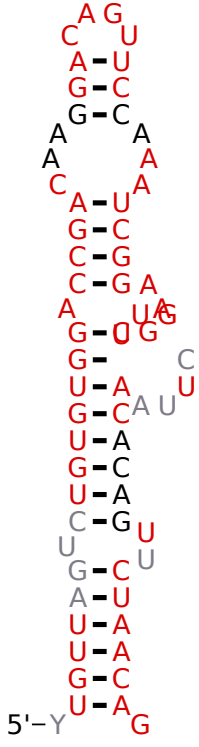
DENV2



DENV3



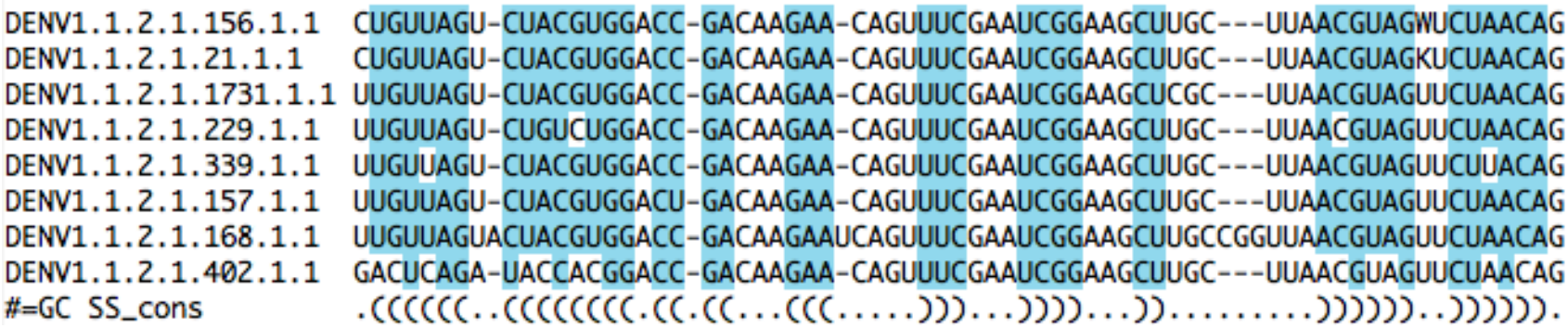
DENV4



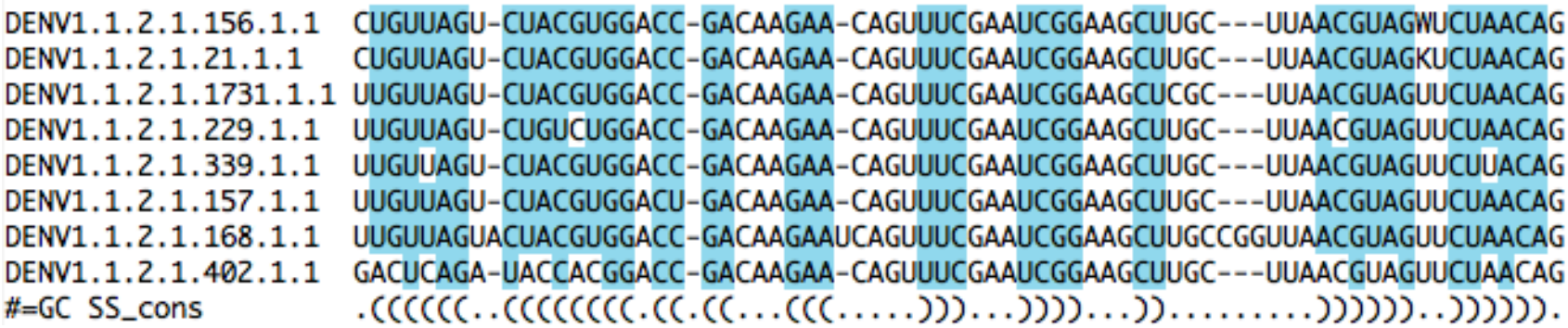
Alternative facts:

- DENV1 folds into another consensus structure
- We need to deconvolute that !

RNAaliSplit objectives



RNAaliSplit objectives



- Find subalignments that have alternative consensus structures
- Identify sequences that do not fit the MSA

RNAaliSplit objectives

```
DENV1.1.2.1.156.1.1   CUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGWUCUAACAG
DENV1.1.2.1.21.1.1    CUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGKUCUAACAG
DENV1.1.2.1.1731.1.1 UUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUCGC---UUAACGUAGUUCUAACAG
DENV1.1.2.1.229.1.1   UUGUUAGU-CUGUCUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
DENV1.1.2.1.339.1.1   UUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUUACAG
DENV1.1.2.1.157.1.1   UUGUUAGU-CUACGUGGACU-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
DENV1.1.2.1.168.1.1   UUGUUAGUACUACGUGGACC-GACAAGAAUCAGUUUCGAAUCGGAAGCUUGCCGGUUAACGUAGUUCUAACAG
DENV1.1.2.1.402.1.1   GACUCAGA-UACCACGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
#=GC SS_cons          .((((((..(((((((((.((.((...(((.....)))...))))...)).........))))))..)))))).
```

- Find subalignments that have alternative consensus structures
- Identify sequences that do not fit the MSA

Structure conservation

Covariation

RNAaliSplit objectives

```
DENV1.1.2.1.156.1.1   CUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGWUCUAACAG
DENV1.1.2.1.21.1.1    CUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGKUCUAACAG
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DENV1.1.2.1.157.1.1   UUGUUAGU-CUACGUGGACU-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
DENV1.1.2.1.168.1.1   UUGUUAGUACUACGUGGACC-GACAAGAAUCAGUUUCGAAUCGGAAGCUUGCCGGUUAACGUAGUUCUAACAG
DENV1.1.2.1.402.1.1   GACUCAGA-UACCACGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
#=GC SS_cons          .((((((..(((((((((.((.((...(((.....)))...))))...)).........))))))..))))))).
```

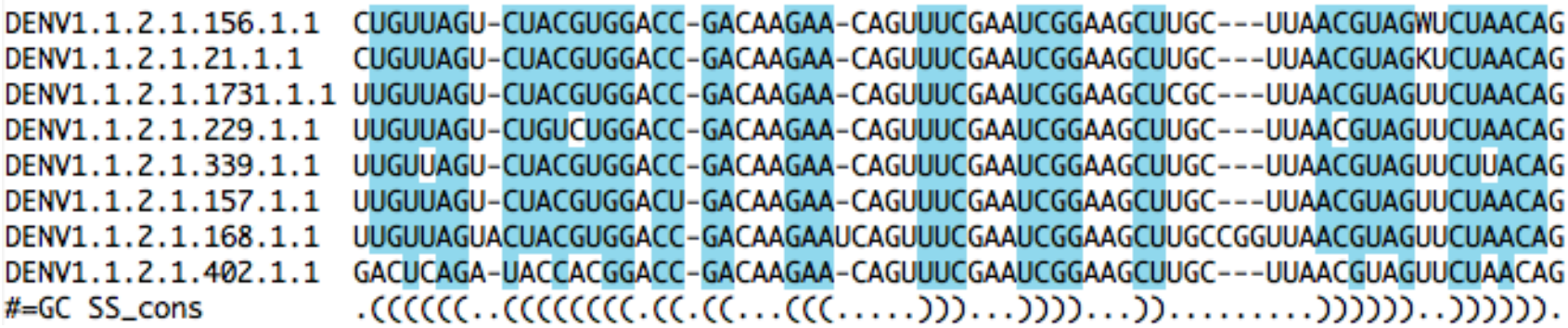
- Find subalignments that have alternative consensus structures
- Identify sequences that do not fit the MSA

Structure conservation

Covariation

How can we quantify them?

RNAaliSplit objectives



- Find subalignments that have alternative consensus structures
- Identify sequences that do not fit the MSA

Structure conservation

- SCI
- z-score
- mean pairwise identity
- GC content

  RNAz SVM RNA-class probability

Covariation

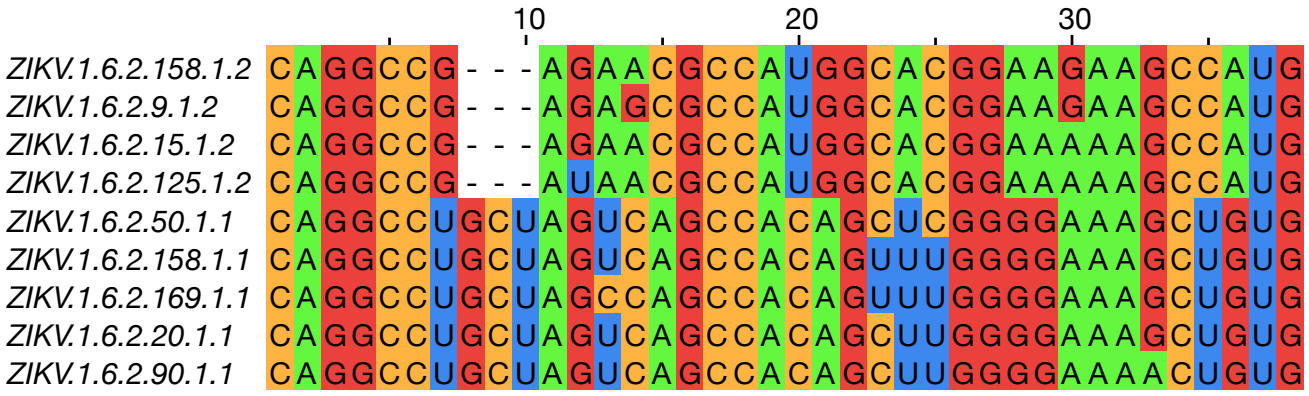
- compensatory mutations

  RNAalifold

- statistical significance

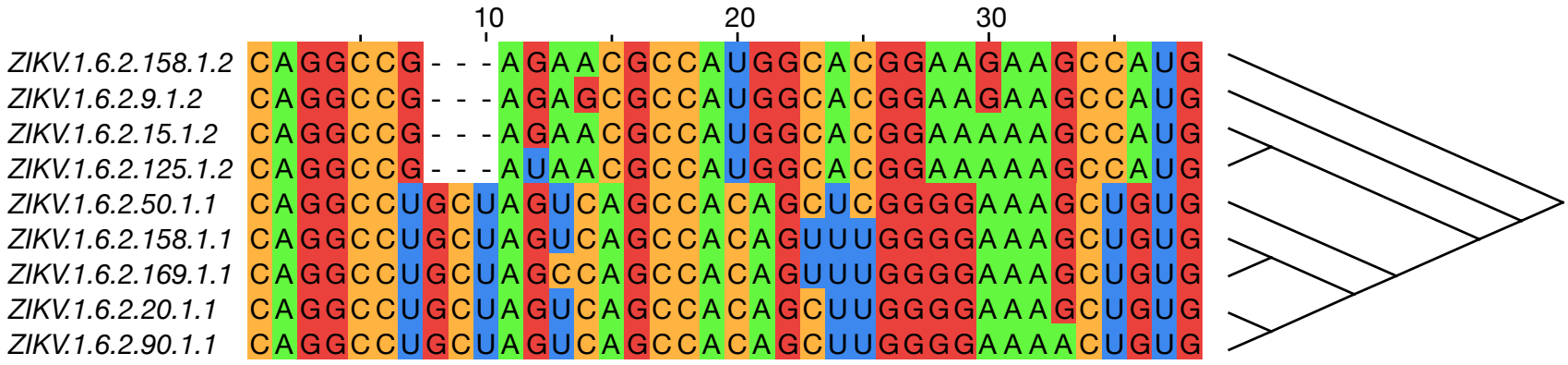
  R-scape

RNAaliSplit workflow



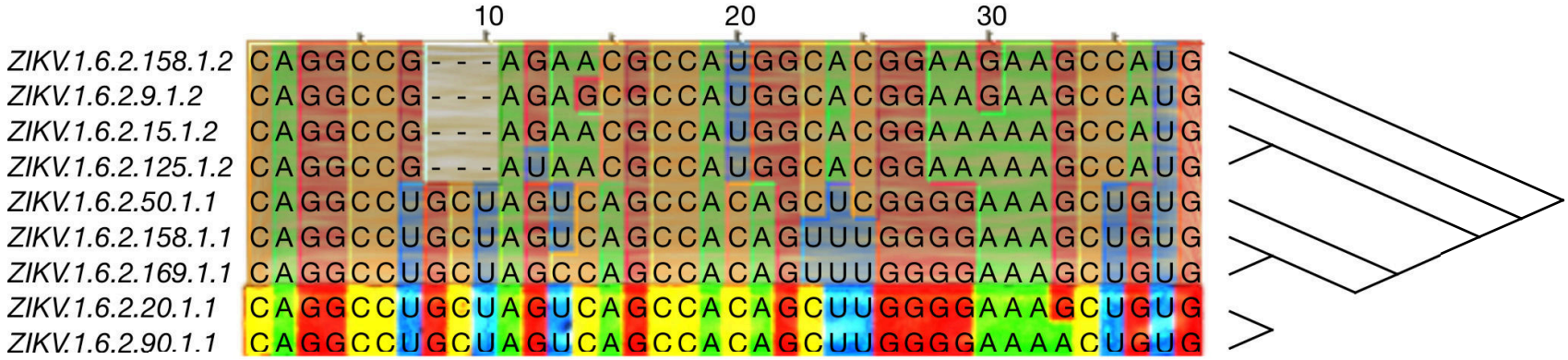

```
SVM prob   z-score SCI      seq consensus structure                                SP  alignment
0.856846   -1.06   0.8613   9 (((((((((.....))).(((((((.......)))))))..)))))) 3  result.aln
-------------------------------------------------------------------------------------------------
0.939812   -1.17   0.8497   7 (((((..........((.(((((.........)))))))..))))) 3  split7.set1.aln
0.036973   -0.65   0.9671   2 (((((..(((....)))(((((((.....)))))))..))))) 0  split7.set2.aln
0.051212   -0.51   1.0091   2 (((((.(((((((...((.((...)).))...)))).))))))) 0  split8.set1.aln
0.887830   -1.18   0.8409   7 (((((..........((.(((((.........)))))))..))))) 2  split8.set2.aln
0.899553   -1.43   0.9442   5 (((((..........((.(((((.........)))))))..))))) 2  split9.set1.aln
0.063624   -0.56   0.9789   4 (((((.(((((((...((.((...)).))...)))).))))))) 0  split9.set2.aln
...
```

RNAaliSplit workflow



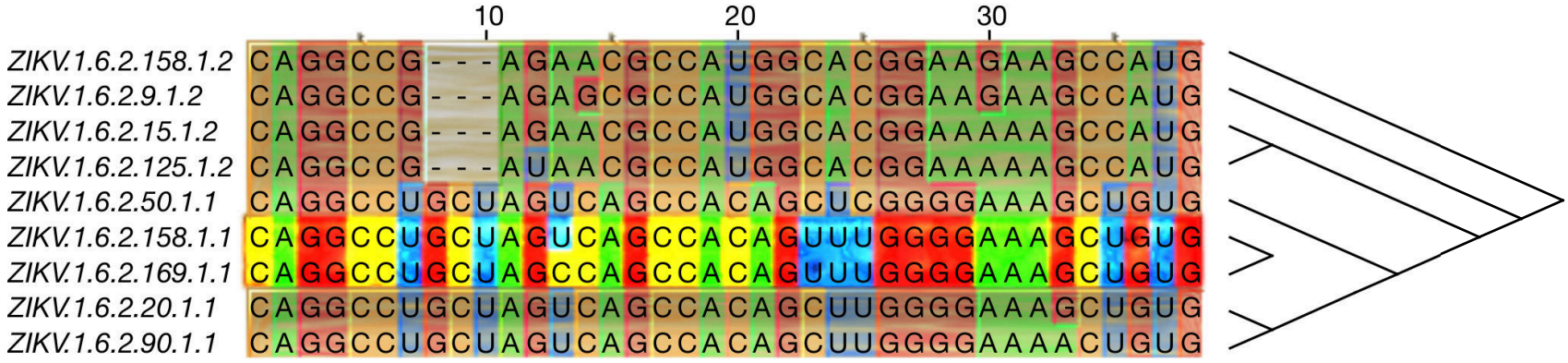

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RNAaliSplit workflow



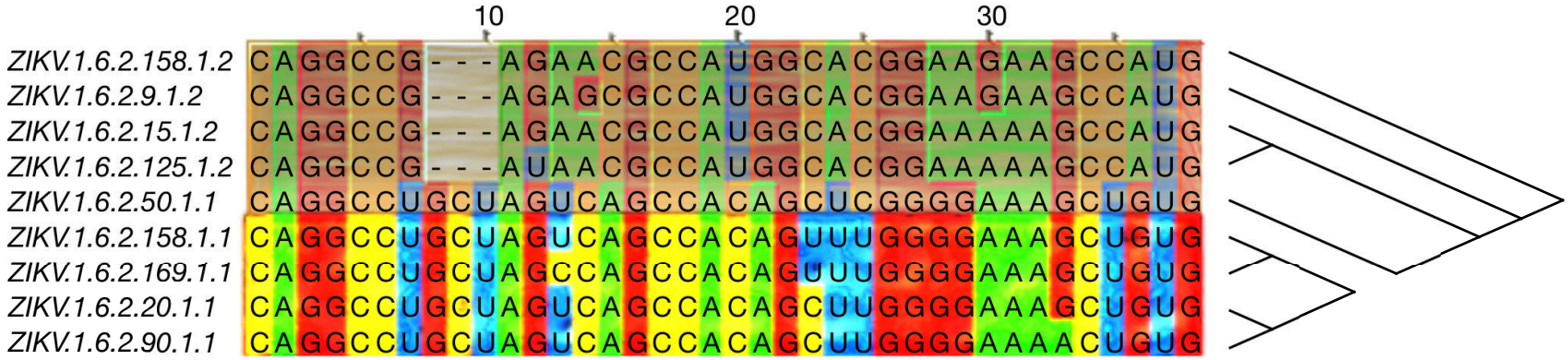

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0.036973   -0.65   0.9671   2 (((((..(((....)))(((((((.....)))))))..))))) 0  split7.set2.aln
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RNAaliSplit workflow



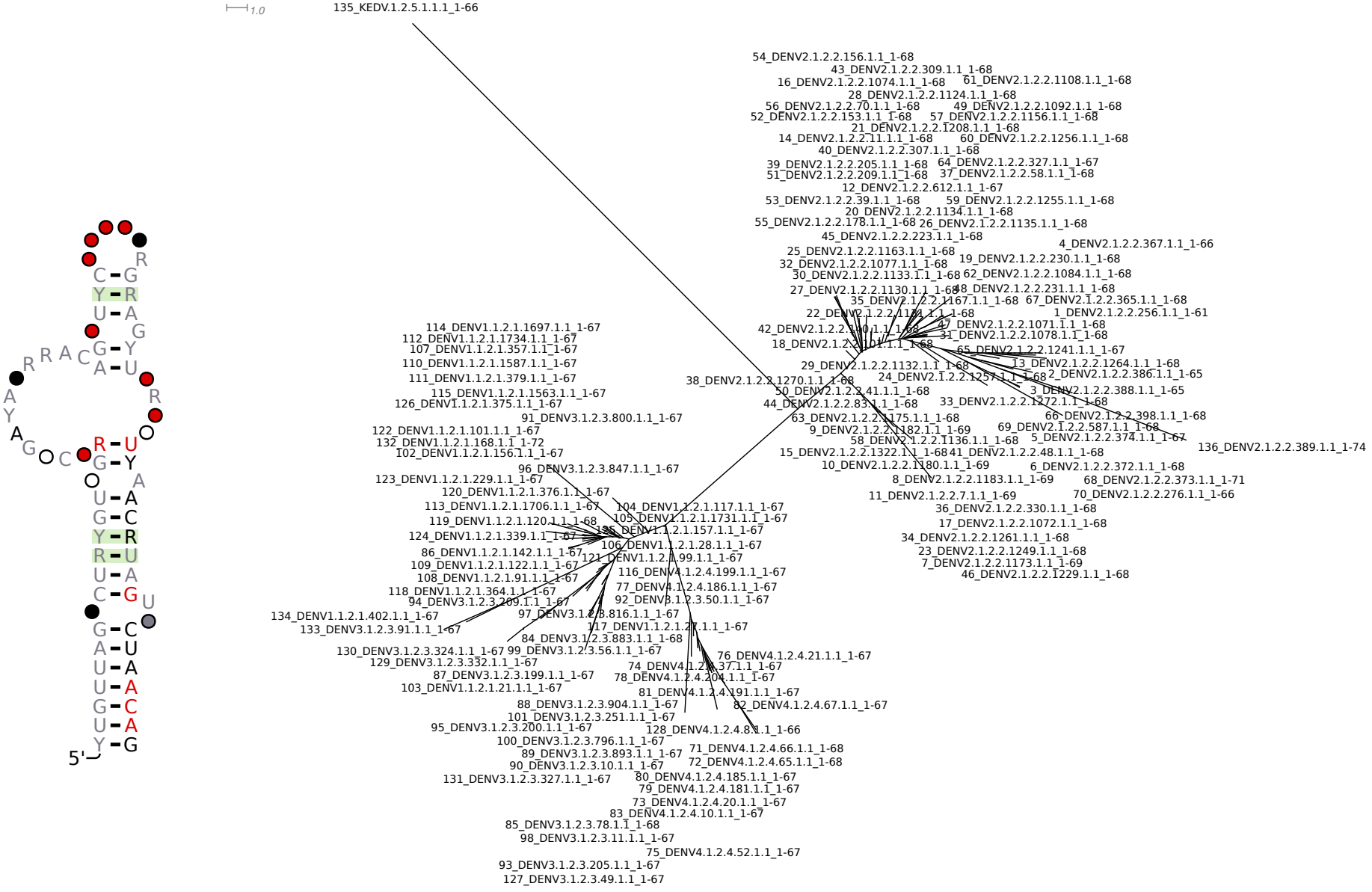

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RNAaliSplit workflow




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...
```

Dengue virus SLA



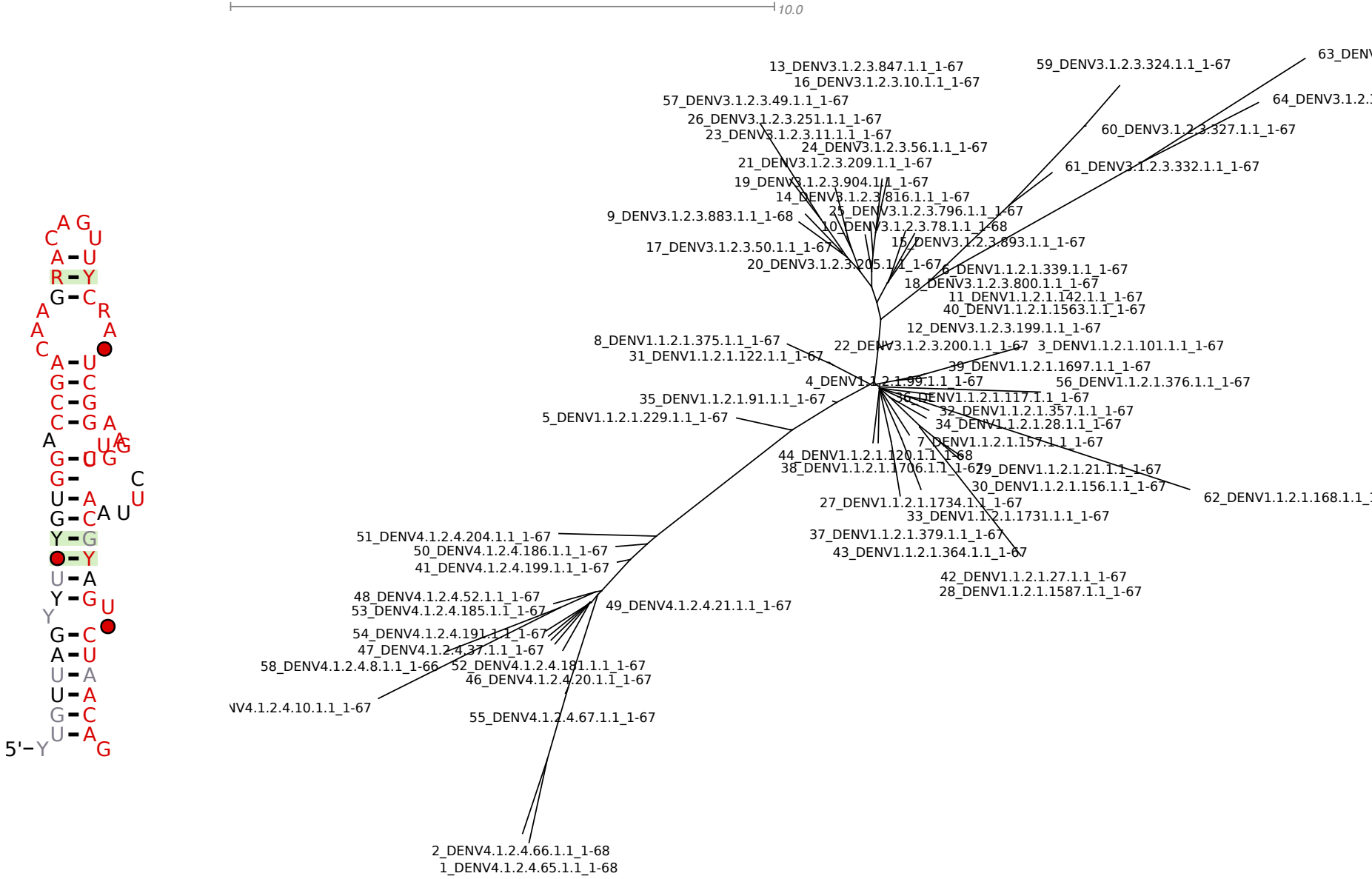
NJ tree; 146 pairwise different DENVG SLA sequences

Dengue virus SLA



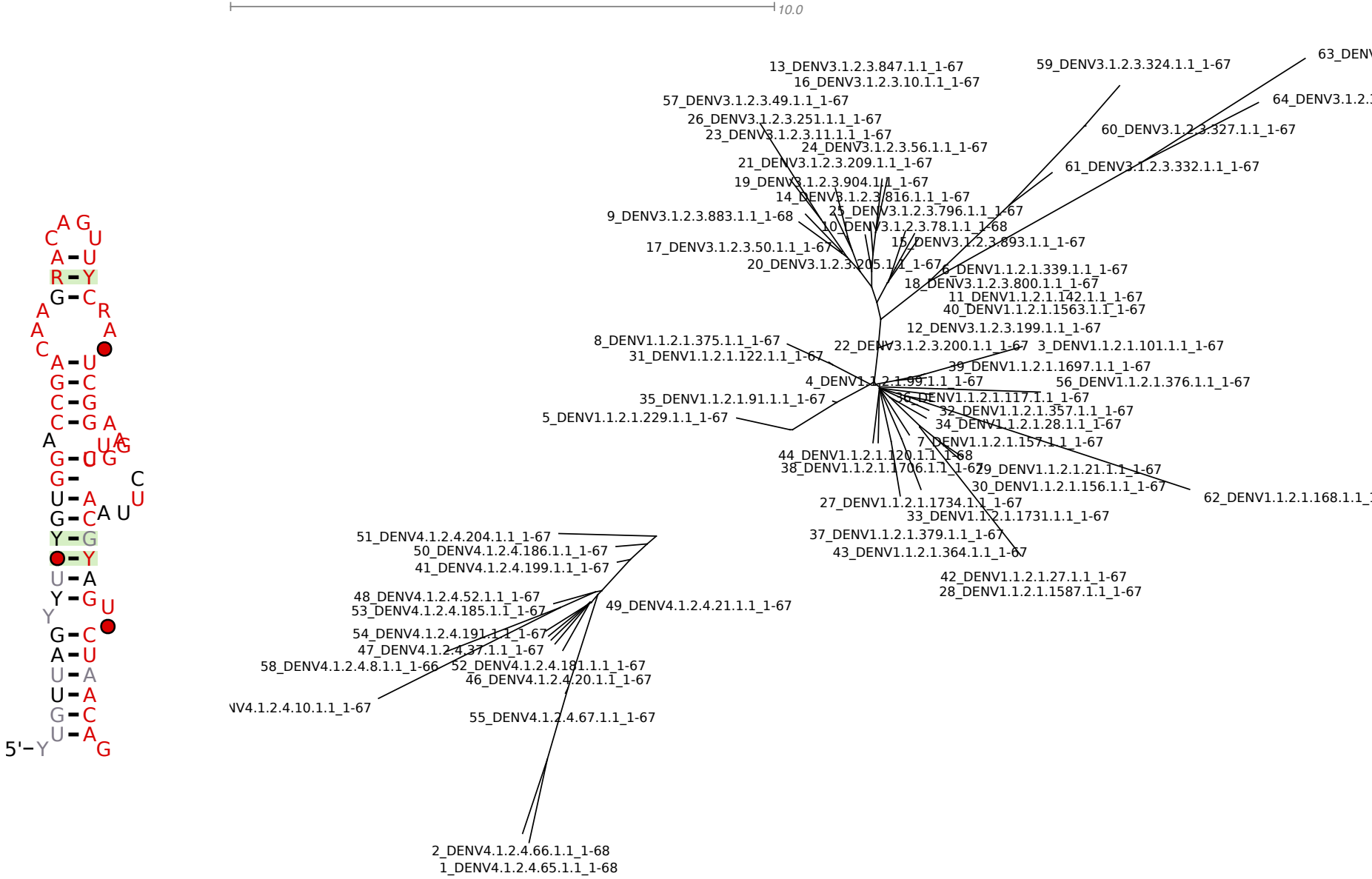
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Dengue virus SLA



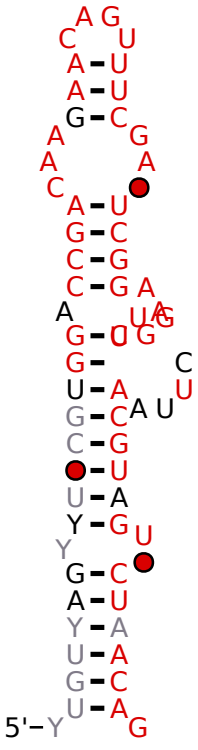
NJ tree; 64 pairwise different DENV1,3,4 SLA sequences

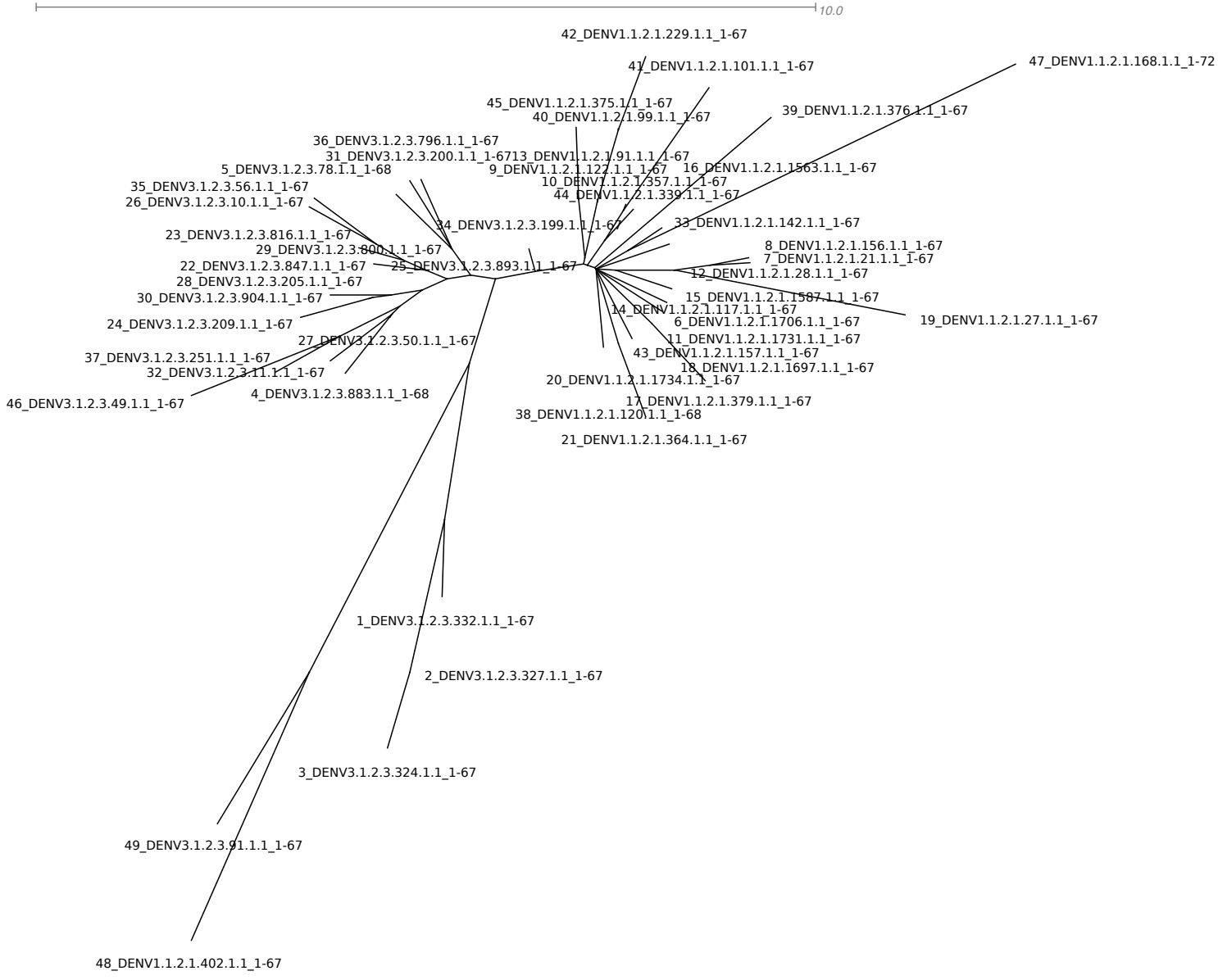
Dengue virus SLA



NJ tree; 64 pairwise different DENV1,3,4 SLA sequences

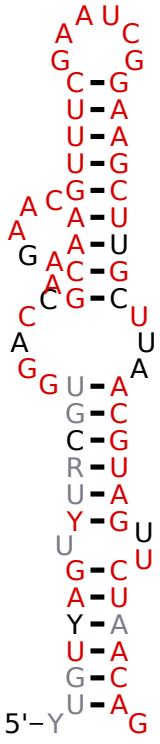
Dengue virus SLA

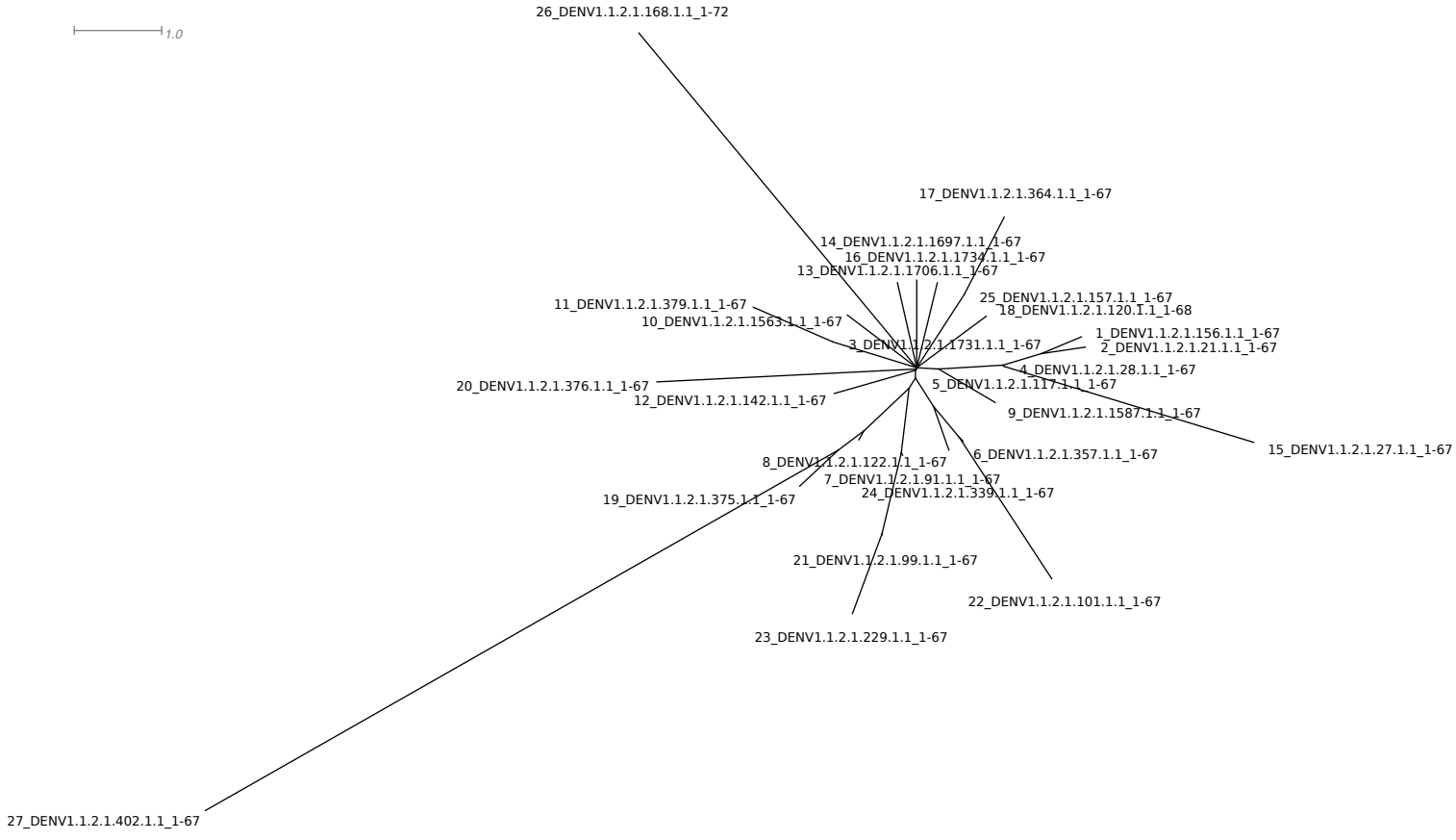




NJ tree; 49 pairwise different DENV1,3 SLA sequences

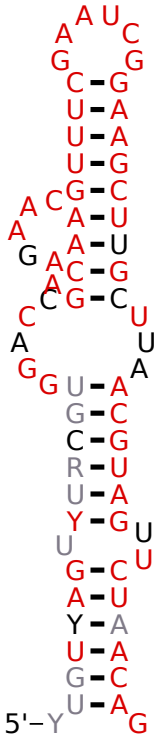
Dengue virus SLA

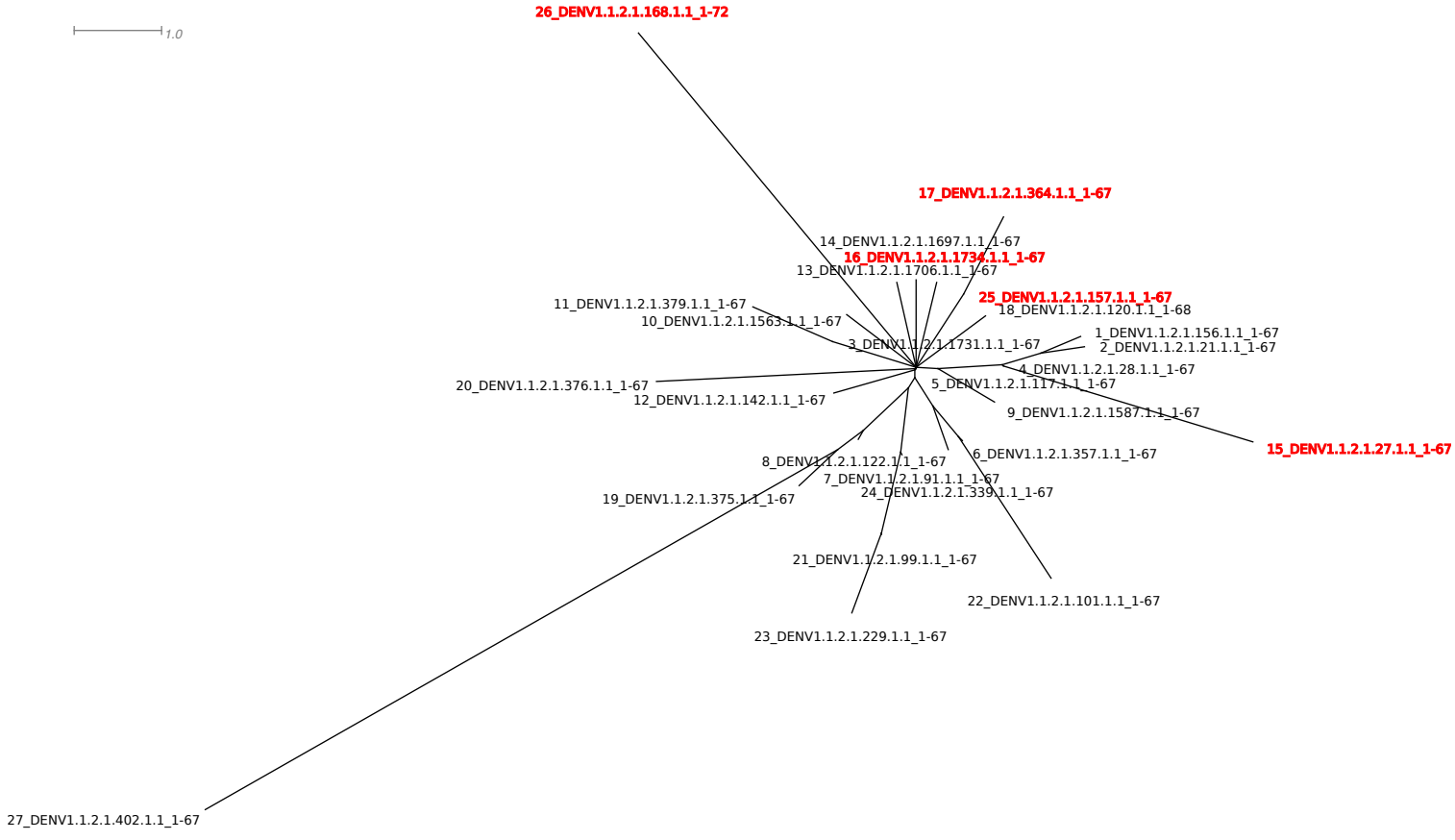




NJ tree; 27 pairwise different DENV1 SLA sequences

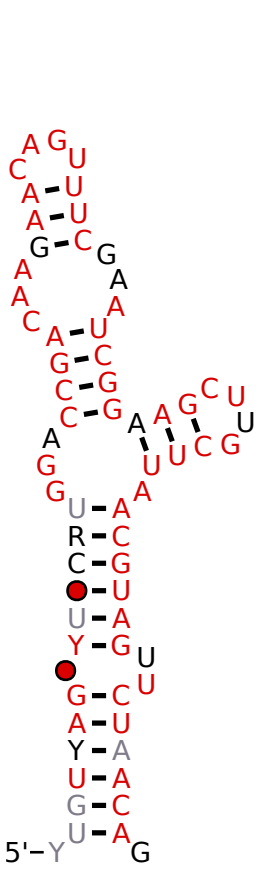
Dengue virus SLA

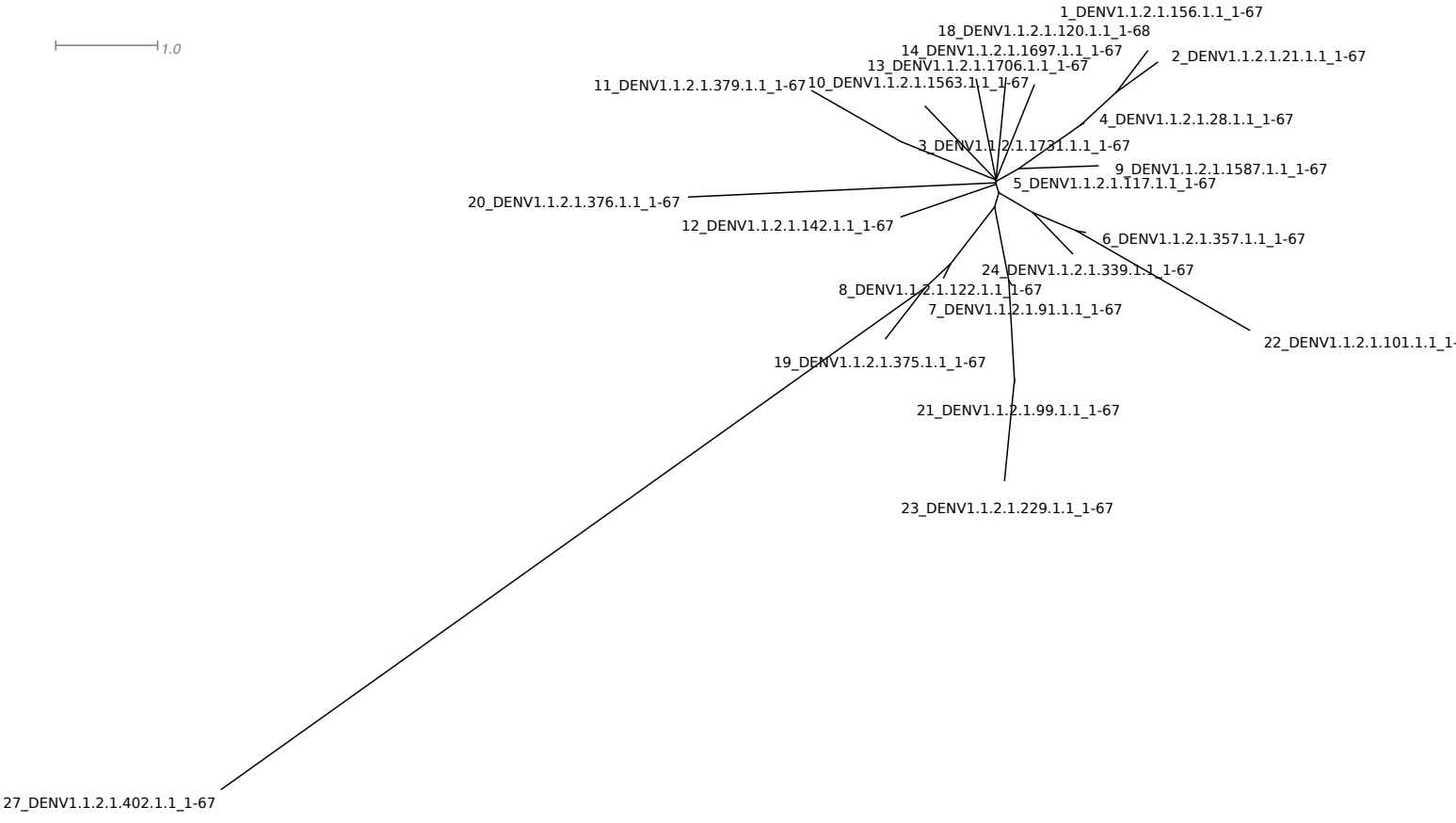




NJ tree; 27 pairwise different DENV1 SLA sequences

Dengue virus SLA





NJ tree; 22 pairwise different DENV1 SLA sequences

Summary

Flavivirus SLA elements are

- required for FV replication, mediating (-)gRNA synthesis by RdRp
- conserved among Mosquito-bourne and Tick-bourne FV

Summary

Flavivirus SLA elements are

- required for FV replication, mediating (-)gRNA synthesis by RdRp
- conserved among Mosquito-bourne and Tick-bourne FV

RNAaliSplit is

- a heuristic approach for splitting MSA into sets that have alternative consensus structures
- a means for cleaning MSA before consructing covariance models
- unversally applicable to moderate-sized RNA MSA
- implemented in a modular, object-oriented way based on the Perl Moose framework
- available soon as a stand-alone Perl Module

Acknowledgments

Collaborators
Andrea Tanzer
Roman Ochsenreiter
Ivo L. Hofacker

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Der Wissenschaftsfonds.
SFB RNA regulation of the transcriptome (FWF-F43)

universität wien
MEDIZINISCHE UNIVERSITÄT WIEN

VIENNA RNA BIOLOGY
DK RNA Biology (FWF-W 1207)

Thank you!